

Dynamic modeling of personalized time-varying treatment effect in mobile health data

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Outline

- 1 Introduction and Motivation
- 2 Hierarchical Bayesian State-Space Model
- 3 Posterior Inference
- 4 Prediction and Online Learning
- 5 Simulation Study
- 6 Conclusion

Intensive Longitudinal Data and the ARC Study

- Smartphones and wearables enable **repeated, fine-grained** measurement of mental health in everyday life.

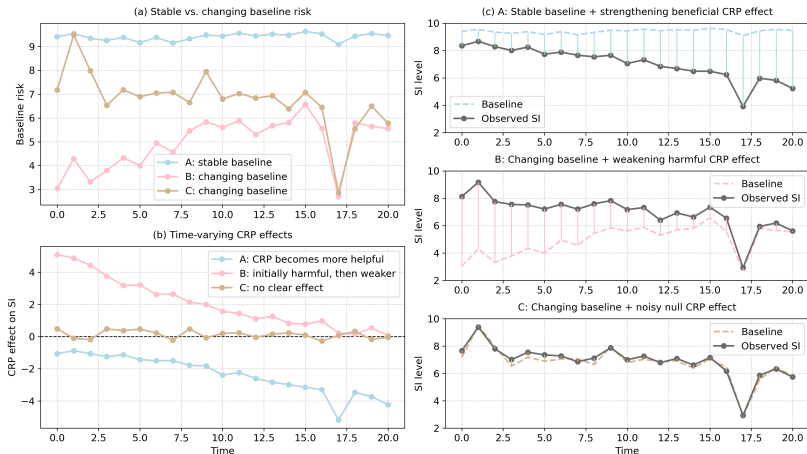
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- **Advancing Resilience and Coping (ARC) study**: a 4-week intensive longitudinal study for individualized suicide prevention
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 - Passive sensing: GPS, phone usage, Fitbit activity & sleep
- **Observed outcome**: Suicidal ideation (SI) severity
- **Treatment of interest**: Crisis Response Plan (CRP) usage

The Challenge: Distinct Mechanisms, Similar Observations



Key point: Superficially similar observed trajectories may arise from distinct underlying mechanisms, requiring different intervention strategies.

Statistical Challenges

- 1 Within-person **temporal dependence**

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- ④ **Latent subgroups** (who benefits? who doesn't? Stable?)

Existing Approaches and Their Limitations

Approach	Strength	Limitation
GLMMs (mixed-effects)	heterogeneity through random intercepts and slopes	static random effects; no meaningful response subgroups

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Hierarchical Bayesian models	information sharing; uncertainty quantification	static subject-specific parameters

Each of them addresses part of the problem, but none combines all the features we need.

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No existing framework jointly integrates **nonstationary latent state evolution**, **individualized time-varying treatment effects**, **latent subgroup discovery**, and **principled information sharing** across individuals.

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Fundamental Tradeoff

Population-level stability $\longleftrightarrow$ Individual-level flexibility

Our goal: balance these via a *hierarchical Bayesian state-space model*

Proposed Model: Observation Equation

For individual i at time t :

$$y_{i,t} = \underbrace{\alpha_{i,t}}_{\text{baseline risk}} + \underbrace{\beta_{i,t} x_{i,t}}_{\text{time-varying effect} \times \text{treatment}} + \underbrace{\mathbf{z}'_{i,t} \boldsymbol{\eta}}_{\text{covariates}} + \epsilon_{i,t}$$

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- $\alpha_{i,t}$: latent baseline risk — **person-specific, time-varying**
- $\beta_{i,t}$: latent treatment effect — **person-specific, time-varying**
- $\epsilon_{i,t} \sim \mathcal{N}(0, \sigma_y^2)$: observation noise

The observed outcome is a **noisy combination** of the latent risk process, treatment response, and contextual factors.

Proposed Model: State Evolution

The latent states evolve as **individual-specific random walks**:

$$\boldsymbol{\theta}_{i,t} = \begin{pmatrix} \alpha_{i,t} \\ \beta_{i,t} \end{pmatrix} = \begin{pmatrix} \alpha_{i,t-1} \\ \beta_{i,t-1} \end{pmatrix} + \boldsymbol{\omega}_{i,t}, \quad \boldsymbol{\omega}_{i,t} \sim \text{N}\left(\mathbf{0}, \begin{pmatrix} \sigma_{\alpha,i}^2 & 0 \\ 0 & \sigma_{\beta,i}^2 \end{pmatrix}\right)$$

- $\sigma_{\alpha,i}^2$ controls how quickly baseline risk changes for person i
- $\sigma_{\beta,i}^2$ controls how quickly treatment effect changes for person i

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The model *adaptively learns* whether each person's trajectory is static or dynamic.

Subject-Level Classification Structure

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- G_i characterizes the **initial treatment effect direction**

$$\beta_{i,0} \mid G_i \sim \underbrace{\mathbb{1}\{G_i = -1\} N(-\mu_1, \tau_1^2)}_{\text{beneficial}} + \underbrace{\mathbb{1}\{G_i = 0\} \delta_0}_{\text{null}} + \underbrace{\mathbb{1}\{G_i = 1\} N(\mu_2, \tau_2^2)}_{\text{harmful}}$$

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- $(D_{\alpha,i}, D_{\beta,i})$ characterize the **temporal dynamics**

$$\sigma_{\alpha,i}^2 \mid D_{\alpha,i} \sim \underbrace{(1 - D_{\alpha,i}) \delta_0}_{\text{static}} + \underbrace{D_{\alpha,i} \text{Exp}(\lambda_{\alpha})}_{\text{time-varying}}$$

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Subject-Level Classification Structure

A 3×2 subgroup structure focusing on $(G_i, D_{\beta,i})$ captures clinically meaningful treatment-response patterns:

Table 1: Subject-level subgroup structure and clinical interpretation.

G_i	$D_{\beta,i}$	Interpretation	Clinical Implication	Prior mean prob.
0	0	No effect (static)	Consider alternative strategies	0.25
0	1	No effect (dynamic)	Monitor closely before changing	0.25
-1	0	Beneficial (static)	Continue treatment	0.125
-1	1	Beneficial (dynamic)	Continue with close monitoring	0.125
+1	0	Harmful (static)	Consider discontinuation	0.125
+1	1	Harmful (dynamic)	Monitor and adjust	0.125

- $G_i \in \{-1, 0, +1\}$: direction of initial treatment effect
- $D_{\beta,i} \in \{0, 1\}$: whether treatment effect evolves over time
- Subgroup membership is **learned from the data**

Graphical Model Overview

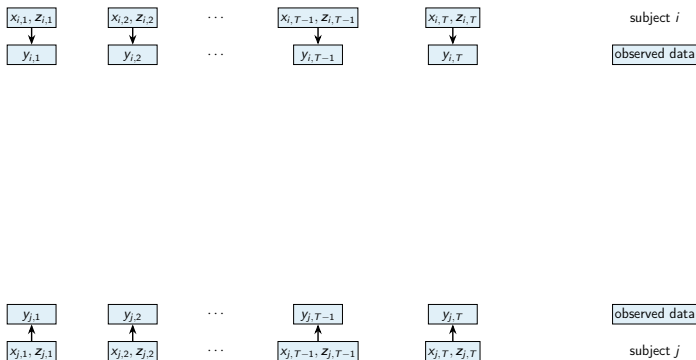


Figure 1: Hierarchical Bayesian state-space model: observed data are linked to time-varying latent states within each individual, while a higher-level classification structure groups individuals into clinically meaningful subgroups. Global parameters are shared across subjects.

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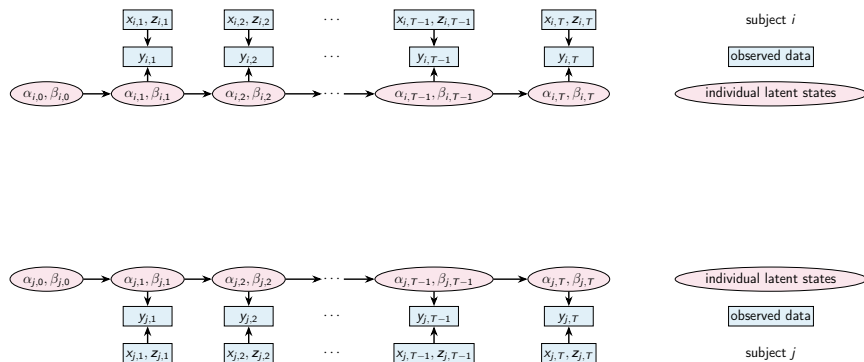


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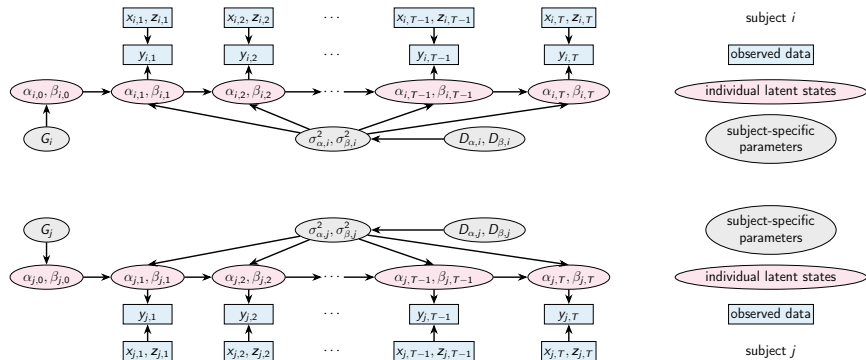


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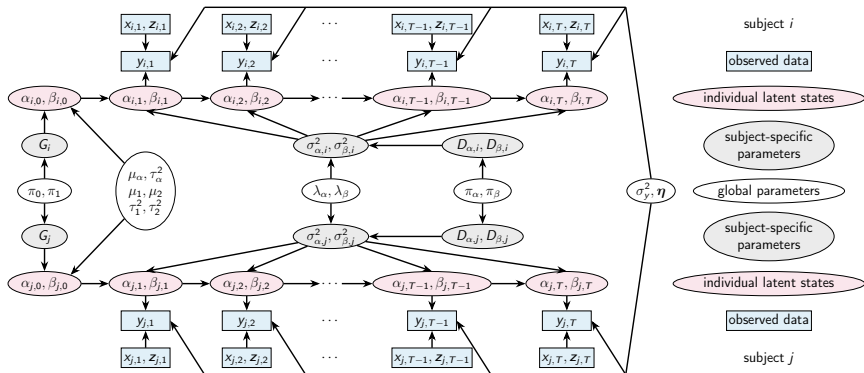


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Posterior Distribution

Target: the joint posterior over all unknowns given the data

$$p\left(\{\boldsymbol{\theta}_{i,0:T}, G_i, D_{\alpha,i}, D_{\beta,i}, \sigma_{\alpha,i}^2, \sigma_{\beta,i}^2\}_{i=1}^N, \boldsymbol{\gamma}, \sigma_y^2, \boldsymbol{\eta} \mid \text{data}\right)$$

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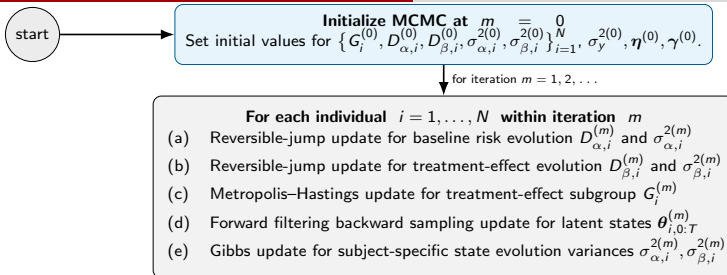
Solution:

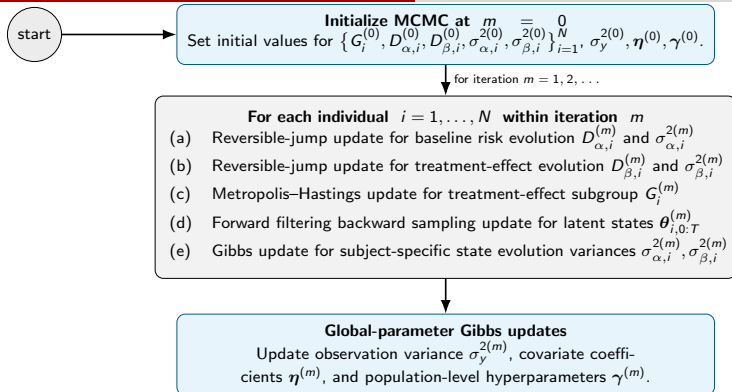
Partially Collapsed Hybrid MCMC that combines reversible-jump, Metropolis–Hastings, forward filtering backward sampling (FFBS) and Gibbs.

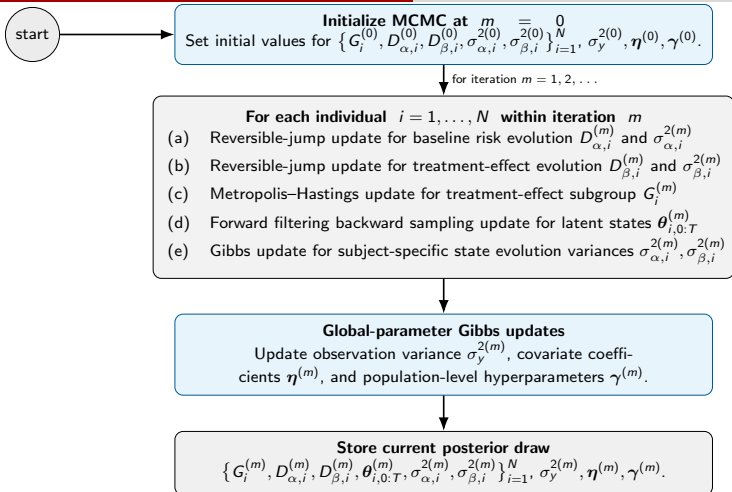
MCMC Sampler: Overview

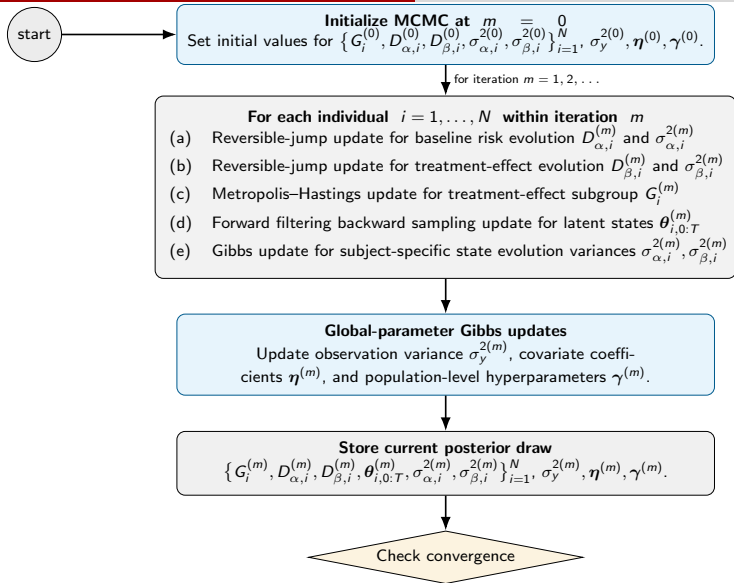
start

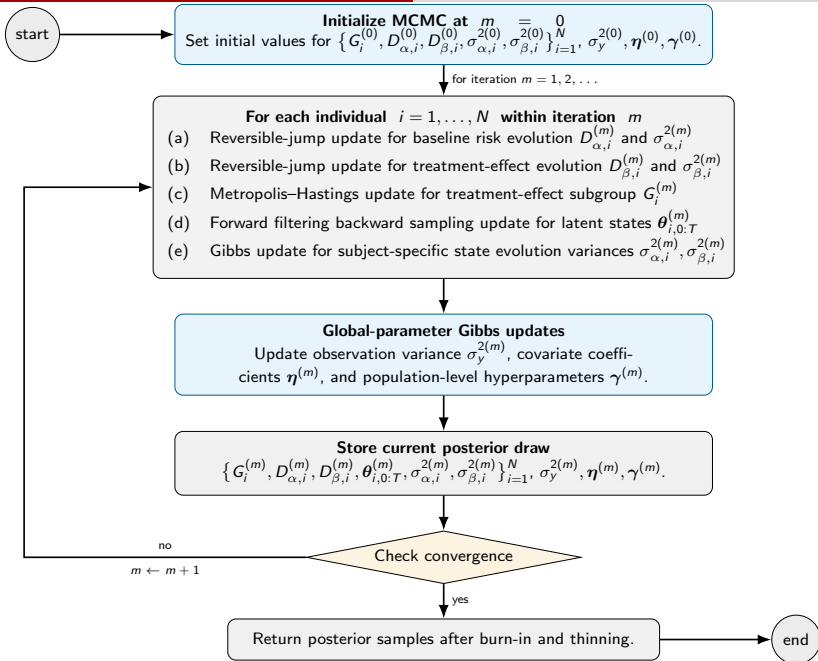
Initialize MCMC at $m = 0$
 Set initial values for $\{G_i^{(0)}, D_{\alpha,i}^{(0)}, D_{\beta,i}^{(0)}, \sigma_{\alpha,i}^{2(0)}, \sigma_{\beta,i}^{2(0)}\}_{i=1}^N, \sigma_y^{2(0)}, \boldsymbol{\eta}^{(0)}, \boldsymbol{\gamma}^{(0)}$.











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Key features:

- Predictive uncertainty grows with forecast horizon q
- **Clinically meaningful:** adapts to each patient's predictability

Online Learning for New Individuals

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Sequential Monte Carlo (SMC) Approach

- 1 **Start from** MCMC posterior draws (equally weighted particles)
- 2 **Augment** each particle with new-subject-specific quantities:
 $(G_{i^*}, D_{\alpha, i^*}, D_{\beta, i^*}, \sigma_{\alpha, i^*}^2, \sigma_{\beta, i^*}^2)$
- 3 **Reweight** particles using new individual's likelihood
- 4 **Resample** if effective sample size is low
- 5 **Perturb** via short MCMC rejuvenation

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Sequential Monte Carlo (SMC) Approach

- 1 MCMC particles
- 2 **Augmentation**
- 3 **Reweighting**
- 4 **Resampling**
- 5 **Perturbation**

Output: posterior subgroup membership probabilities and predictive forecasts for the new patient — **without refitting** the full model.

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- 3 **Online learning**: sequential updating for new individuals (vs. batch MCMC)

100 independent simulation runs for estimation; representative run for forecasting.

Parameter Estimation: Global Parameters

- Overall, the posterior means of global parameters concentrate near their true values.
- Good recovery of the global parameters is encouraging, but our main interest is at the individual level.

Subgroup Classification and Forecasting

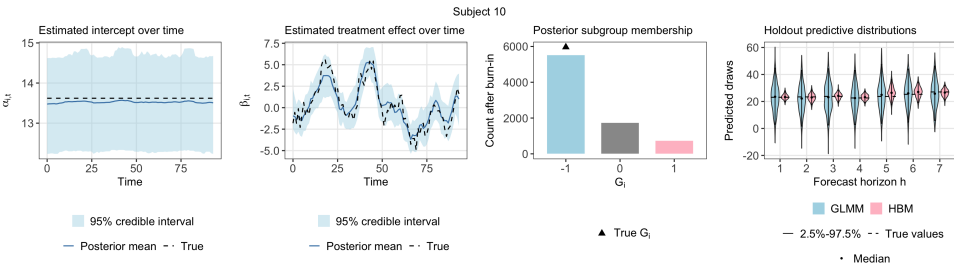


Figure 2: 1 subject: estimated latent trajectories, subgroup classification, and forecasts

- Subgroup classification: uncertain cases are visible through diffuse posterior probabilities, not hidden by a single hard label. Posterior probabilities quantify classification uncertainty.

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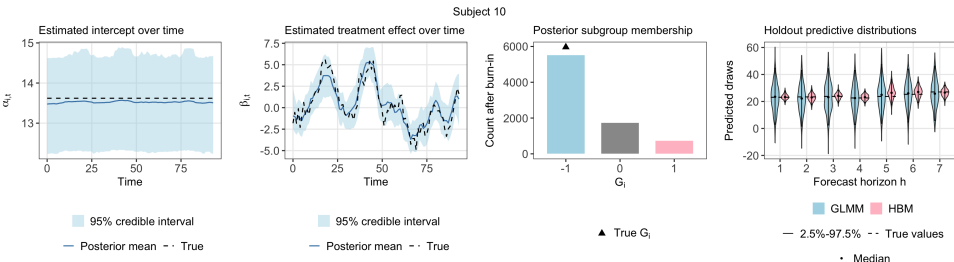


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- Forecasting: proposed model vs. bayesian GLMM

Sequential Updating: SMC vs. Batch MCMC

Subgroup classification:

- SMC closely reproduces batch posterior classifications.
- Discrepancies occur when batch MCMC itself is uncertain.

Computational efficiency: 44.5% reduction in runtime

Table 2: SMC misclassification.

Subject	Indicator
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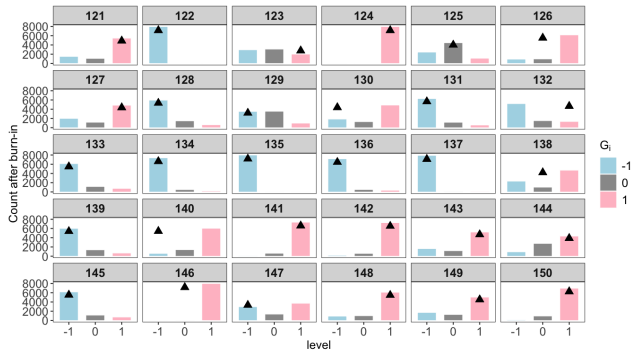


Figure 3: Batch MCMC posterior of G_i for new subjects. $\blacktriangle$ = true values.

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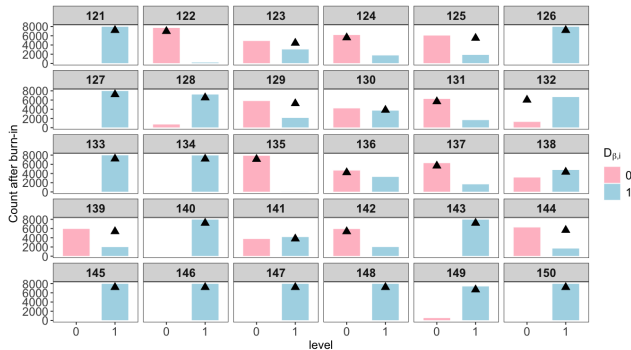


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Broader Impact

This work contributes to **precision medicine** by supporting **individualized, dynamic intervention strategies** that adapt to patients' changing risk profiles over time.

Acknowledgement

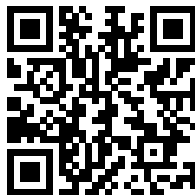
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Thank You!

Questions and comments are welcome.
Open for collaboration.

Scan the QR code to visit my personal website and download these slides.
The manuscript is expected to be available on arXiv in December 2026.

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Limitation and Future Directions

- **Non-Gaussian outcomes:** binary, count, or zero-inflated data commonly arising in psychiatric applications
- **Richer state evolution:** mean reversion, regime switching, or covariate-dependent dynamics
- **Robustness:** missing data and irregular observation times in EMA studies
- **Real-data application:** apply the proposed framework to the ARC study dataset

Parameter Estimation: Global Parameters

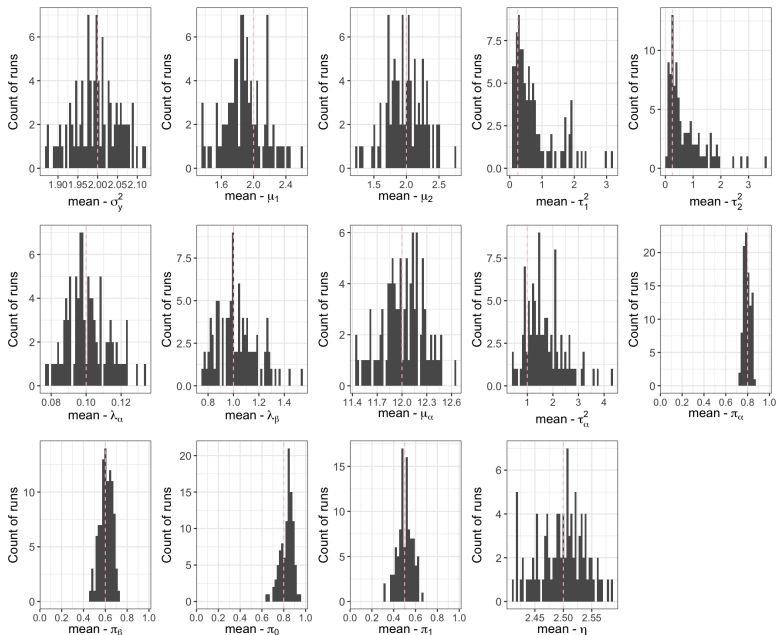


Figure 4: Run-level posterior means of global parameters across 100 independent runs.

Forecasting: Proposed Model vs. Bayesian GLMM

Table 3: Forecasting performance by step (within-subject holdout).

Step	Method	RMSE	MAE	Coverage	Avg. 95% Width
1	HBM	3.05	2.40	0.96	12.54
1	GLMM	15.59	10.98	0.85	41.14
3	HBM	5.16	3.72	0.97	18.41
3	GLMM	16.93	11.51	0.86	41.10
7	HBM	7.88	5.38	0.98	26.48
7	GLMM	15.96	10.64	0.86	41.14

- The proposed model has lower prediction error at each horizon.
- Predictive intervals are adaptive: wider for longer horizons, but not uniformly over-wide.